Two-Stage Product Search Ranker

Hybrid retrieval, a LightGBM cascade and a fine-tuned cross-encoder on the Shopping Queries (ESCI) dataset

Problem

Given a shopping query, rank products from a 482k catalog so the ones that exactly match come first, within a 100 ms budget.

Every labelled query-product pair carries one of four labels:

	label	gain	share
E	Exact: the product matches the query	1.0	43.5%
S	Substitute: close, but some detail differs	0.1	35.1%
C	Complement: goes with what was asked for	0.01	4.5%
I	Irrelevant	0	16.9%

Data

US English queries, split by query so none lands in two splits. The cross-encoders never train on pairs LightGBM learns from.



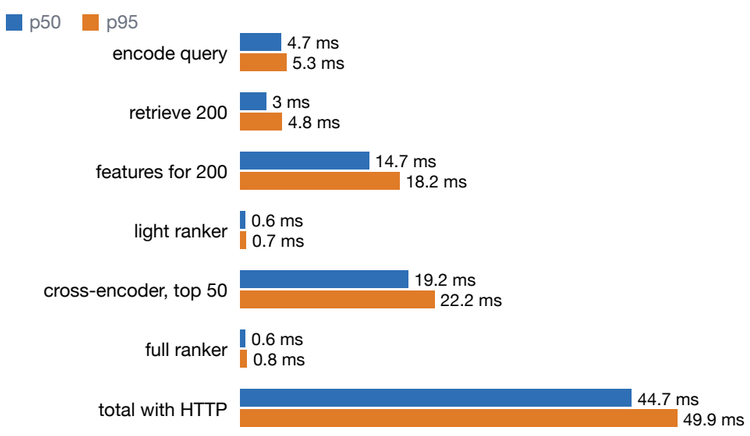
Metrics

$$\text{NDCG@k} = \text{DCG@k} \,/\, \text{ideal DCG@k}, \quad \text{DCG@k} = \Sigma_{i \le k}\, \text{gain}_i \,/\, \log_2(i+1)$$

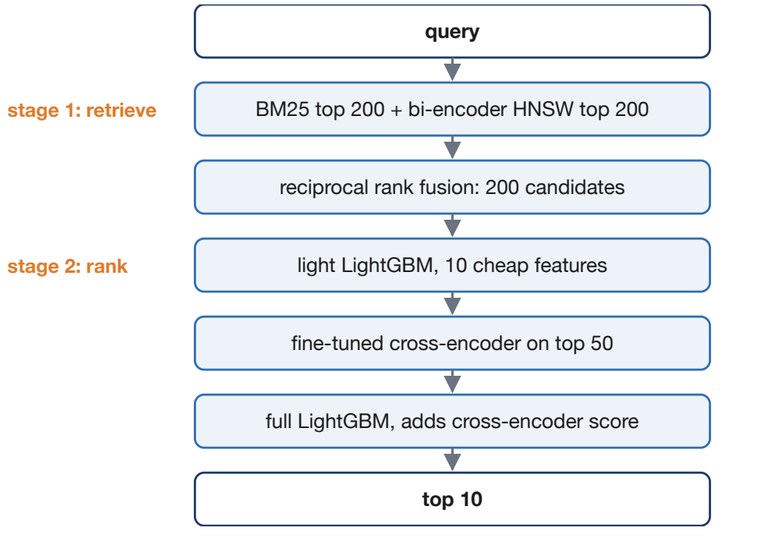
Full-list NDCG scores each query's whole labelled list, the KDD Cup leaderboard metric. **NDCG@10** looks only at what a shopper sees and is much stricter.

Recall@k is the share of Exact products that retrieval finds in its top k of the whole catalog.

Latency, 500 replayed queries

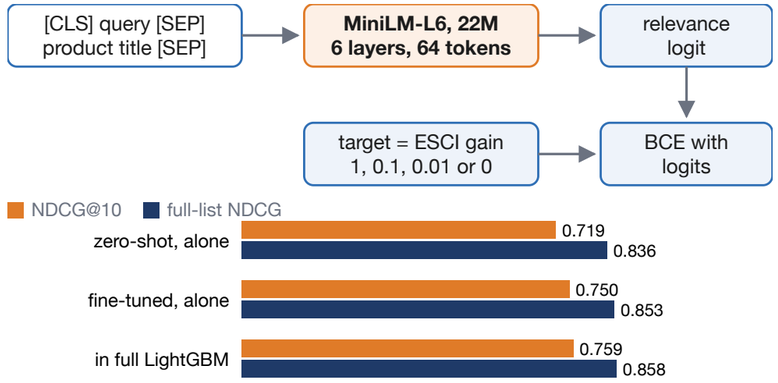


System



Fine-tuning the cross-encoder

An MS MARCO cross-encoder reads query and title together. One epoch on 8.3k ce_train queries, about 5 minutes on a laptop GPU, teaches it ESCI's notion of a match.



End to end over the whole catalog

Each test query searches all 482k products and the top 10 is scored. Products nobody labelled count as 0, so this is a lower bound.

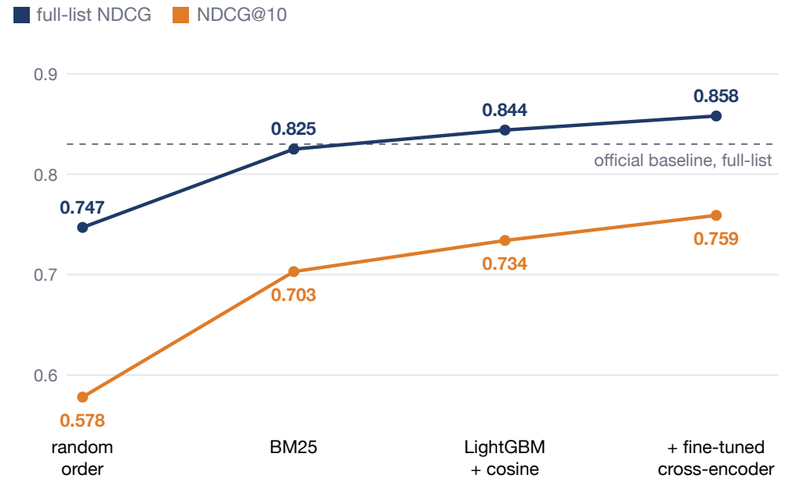


Results on the official test split

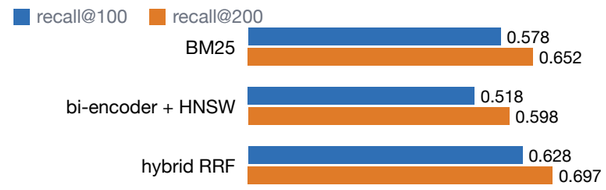
0.858 full-list NDCG

0.759 NDCG@10

45 ms p50 end to end

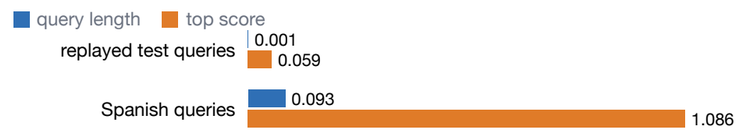


Retrieval recall of Exact products



Drift: watch the output too

PSI against a reference, on query length (input) and the top result's score (output). Above 0.25 means drift.



Takeaways

- Hybrid retrieval finds 5 points more Exact products than BM25 alone, because lexical and dense search miss different products.
- Lexical features stall at 0.725 NDCG@10. Telling Exact from Substitute needs a model that reads both texts together.
- The cascade spends the cross-encoder only on the 50 candidates the cheap ranker likes best.
- Fine-tuning moved NDCG@10 more than any feature, and its score takes 68% of the full ranker's gain importance.